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HTML Canvas Reference

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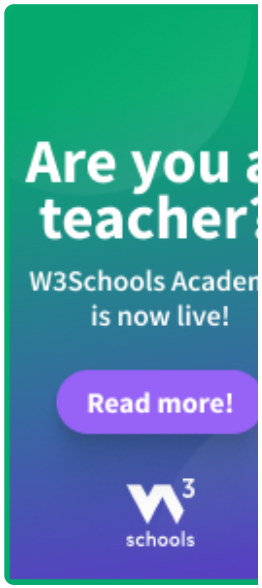
The **<canvas>** element defines a **bitmapped** area in an HTML page.

The **Canvas API** allows JavaScript to **draw graphics** on the canvas.

The Canvas API can draw shapes, lines, curves, boxes, text, and images, with colors, rotations, transparencies, and other pixel manipulations.

You can add a canvas element anywhere in an HTML page with the `<canvas>` tag:

Example



COLOR PICKER

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You can access a `<canvas>` element with the **HTML DOM** method `getElementById()`.

To draw in the canvas you need to create a **2D context** object:

```
const myCanvas = document.getElementById("myCanvas");
const ctx = myCanvas.getContext("2d");
```

Note

The HTML `<canvas>` element itself has no drawing abilities.

You must use JavaScript to draw any graphics.

The `getContext()` method returns an object with tools (methods) for drawing.

Drawing on the Canvas

After you have created a 2D context, you can draw on the canvas.

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Example

```
const myCanvas = document.getElementById("myCanvas");
const ctx = myCanvas.getContext("2d");

ctx.fillRect(20, 20, 150, 100);
```

Try it Yourself »

Using Colors

The **fillStyle** property sets the fill color of the drawing object:

Example

```
const myCanvas = document.getElementById("myCanvas");
const ctx = myCanvas.getContext("2d");

ctx.fillStyle = "red";
ctx.fillRect(20, 20, 150, 100);
```

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You can also create a new `<canvas>` element with the `document.createElement()` method, and add the element to an existing HTML page:

Example

```
const myCanvas = document.createElement("canvas");
document.body.appendChild(myCanvas);
const ctx = myCanvas.getContext("2d");

ctx.fillStyle = "red";
ctx.fillRect(20, 20, 150, 100);
```

Try it Yourself »

Paths

The common way to draw on the canvas is to:

1. Begin a Path - beginPath()
2. Move to a Point - moveTo()
3. Draw in the Path - lineTo()
4. Draw the Path - stroke()

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```
const ctx = canvas.getContext("2d");

ctx.beginPath();
ctx.moveTo(20, 20);
ctx.lineTo(20, 100);
ctx.lineTo(70, 100);
ctx.stroke();
```

Try it Yourself »

Complete Canvas API Reference

This reference covers all Properties and Methods of the getContext("2d") Object, used to draw text, lines, boxes, circles, pictures, and more on the canvas.

Drawing Methods

There are only 3 methods to draw directly on the canvas:

Method	Description

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clearRect()	Clears specified pixels within a rectangle

Path Methods

Method	Description
beginPath()	Begins a new path or resets the current path
closePath()	Adds a line to the path from the current point to the start
isPointInPath()	Returns true if the specified point is in the current path
moveTo()	Moves the path to a point in the canvas (without drawing)
lineTo()	Adds a line to the the path
fill()	Fills the current path
rect()	Adds a rectangle to the path
stroke()	Draws the current path
	Circles and Curves
bezierCurveTo()	Adds a cubic Bézier curve to the path
arc()	Adds an arc/curve (circle, or parts of a circle) to the path
arcTo()	Adds an arc/curve between two tangents to the path

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Text

Method/Prop	Description
direction	Sets or returns the direction used to draw text
fillText()	Draws "filled" text on the canvas
font	Sets or returns the font properties for text content
measureText()	Returns an object that contains the width of the specified text
strokeText()	Draws text on the canvas
textAlign	Sets or returns the alignment for text content
textBaseline	Sets or returns the text baseline used when drawing text

Colors, Styles, and Shadows

Method/Property	Description
addColorStop()	Specifies the colors and stop positions in a gradient object

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createRadialGradient()	Creates a radial/circular gradient (to use on canvas content)
fillStyle	Sets or returns the color, gradient, or pattern used to fill the drawing
lineCap	Sets or returns the style of the end caps for a line
lineJoin	Sets or returns the type of corner created, when two lines meet
lineWidth	Sets or returns the current line width
miterLimit	Sets or returns the maximum miter length
shadowBlur	Sets or returns the blur level for shadows
shadowColor	Sets or returns the color to use for shadows
shadowOffsetX	Sets or returns the horizontal distance of the shadow from the shape
shadowOffsetY	Sets or returns the vertical distance of the shadow from the shape
strokeStyle	Sets or returns the color, gradient, or pattern used for strokes

Transformations

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rotate()	Rotates the current drawing
translate()	Remaps the (0,0) position on the canvas
transform()	Replaces the current transformation matrix for the drawing
setTransform()	Resets the current transform to the identity matrix. Then runs transform()

Image Drawing

Method	Description
drawImage()	Draws an image, canvas, or video onto the canvas

The ImageData Object / Pixel Manipulation

Method/Property	Description
createImageData()	Creates a new, blank ImageData object
getImageData()	Returns an ImageData object that copies the pixel data for the specified rectangle on a canvas

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ImageData.width	Returns the width of an ImageData object
putImageData()	Puts the image data (from a specified ImageData object) back onto the canvas

Compositing

Property	Description
globalAlpha	Sets or returns the current alpha or transparency value of the drawing
globalCompositeOperation	Sets or returns how a new image are drawn onto an existing image

Other Methods

Method	Description
clip()	Clips a region of any shape and size from the original canvas
save()	Saves the state of the current drawing context and all its attributes

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getContext()
toDataURL()

Standard Properties and Events

The canvas object also supports the standard properties and events.

Related Pages

Canvas tutorial: Canvas Tutorial

HTML tutorial: HTML5 Canvas

HTML reference: HTML <canvas> tag

Browser Support

The `<canvas>` element is an HTML5 standard (2014).

`Canvas API` is supported in all modern browsers:

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